

Week 6 - Part 3 - Osprey1st College Algebra - Summer 2024

Section covered: 9.6, 10.1, and 10.2

Solving an Exponential or Logarithmic Equation

To solve an exponential or logarithmic equation, change the given equation into one of the following forms, where a and b are real numbers, $a > 0$ and $a \neq 1$, and follow the guidelines.

1. $a^{f(x)} = a^{g(x)}$

The given equation is equivalent to the equation $f(x) = g(x)$. Solve algebraically.

2. $a^{f(x)} = b$

Solve by taking natural (or common) logarithms on each side $\ln a^{f(x)} = \ln b$ and then bring $f(x)$ in front of the logarithm, $f(x) \ln a = \ln b$.

3. $\log_a f(x) = b$

Solve by changing to exponential form $a^b = f(x)$. Check that each proposed solution is in the domain.

4. $\log_a f(x) = \log_a g(x)$

The given equation is equivalent to the equation $f(x) = g(x)$. Solve algebraically. Check that each proposed solution is in the domain.

Logarithmic Equations

Recall that the domain of $y = \log_a x$ is $(0, \infty)$. For this reason, it is always necessary to check that proposed solutions of a logarithmic equation result in logarithms of positive numbers in the original equation.

Problem 3. * Solve each equation. Give exact values.

(a) $8 + \log_5(-4x) = 10$

(b) $6 \log_5(x + 27) = 14$

$$(c) \log_2(x+5) = -\log_2(x+3) + 3$$

$$(d) \log_4(x+6) = 1 + \log_4(x+3)$$

$$(e) \log_5(x-7) = \log_5(x+1) + 1$$

Compound interest is interest computed on your original investment as well as on any accumulated interest.

Suppose a sum of money, called the **principal**, P , is invested at an annual percentage rate r , in decimal form, compounded once per year. Because the interest is added to the principal at year's end, the accumulated value, A , is

$$A(1) = P + Pr = P(1 + r).$$

The accumulated value after 2, 3, and 4 years is

$$\begin{aligned} A(2) &= P(1 + r) + P(1 + r)r = P(1 + r)(1 + r) = P(1 + r)^2 \\ A(3) &= P(1 + r)^2 + P(1 + r)^2r = P(1 + r)^2(1 + r) = P(1 + r)^3 \\ A(4) &= P(1 + r)^3 + P(1 + r)^3r = P(1 + r)^3(1 + r) = P(1 + r)^4 \end{aligned}$$

If there are n compounding periods per year, in each time period the interest rate is rn and there are nt time periods in t years. This results in the following formula for the balance, A , after t years:

$$A(t) = \left(1 + \frac{r}{n}\right)^{nt}$$

Compound Interest

After t years, the balance, A , in an account with principal (amount deposited) P and annual interest rate r (in decimal form) is given by the following formulas:

$$A = P \left(1 + \frac{r}{n}\right)^{nt}$$

Problem 4. Suppose \$2500 is deposited in an account paying 6% interest per year compounded semiannually (twice per year).

(a) Find the amount in the account after 10 yr with no withdrawals.

(b) How much interest is earned over the 10-yr period?

Problem 5. * Salma needs 7110 for a future project. She can invest \$6000 now at an annual rate of 3%, compounded monthly. Assuming that no withdrawals are made, how long will it take for her to have enough money for her project? Do not round any intermediate computations, and round your answer to the nearest hundredth.

Problem 6. * Suppose that \$4000 is placed in a savings account at an annual rate of 4.6%, compounded quarterly. Assuming that no withdrawals are made, how long will it take for the account to grow to \$5572?

The more often interest is compounded within a given time period, the more interest will be earned. Surprisingly, however, there is a limit on the amount of interest, no matter how often it is compounded.

n	$A = \left(1 + \frac{1}{n}\right)^n$
1	2
2	2.25
5	2.48832
10	2.59374
100	2.70481
1000	2.71692
10,000	2.71815
1,000,000	2.71828

Suppose that \$1 is invested at 100% interest per year, compounded n times per year. According to the formula (with $P = 1$ and $r = 1$), the compound amount at the end of 1 yr will be

$$A = \left(1 + \frac{1}{n}\right)^n$$

The table suggests that as n increases, the value of $\left(1 + \frac{1}{n}\right)^n$ gets closer and closer to some fixed number. This is indeed the case. This fixed number is called e .

Value of e

$$e \approx 2.718281828459045$$

If we let the number of compounding periods increases infinitely, we call this type of compounding, **continuous compounding**. As shown above, the amount of interest earned increases with the frequency of compounding, but the value of the expression $\left(1 + \frac{1}{n}\right)^n$ approaches e as n gets larger. Consequently, the formula for compound interest approaches a limit as well, called the compound amount from continuous compounding.

Continuous Compounding

If P dollars are deposited at a rate of interest r compounded continuously for t years, then the compound amount A in dollars on deposit is given by the following formula.

$$A = Pe^{rt}$$

Formulas for Compound Interest

After t years, the balance, A , in an account with principal (amount deposited) P and annual interest rate r (in decimal form) is given by the following formulas:

1. For n compounding periods per year: $A = P \left(1 + \frac{r}{n}\right)^{nt}$
2. For continuous compounding: $A = Pe^{rt}$.

The Exponential Growth/Continuous Compounding

In many situations that occur in ecology, biology, economics, and the social sciences, a quantity changes at a rate proportional to the amount present. In such cases, the amount present at time t is a special function of t called an **exponential growth or decay function**.

Exponential Growth or Decay Function

Let y_0 be the amount or number present at time $t = 0$. Then, under certain conditions, the amount present at any time t is modeled by

$$y = y_0 e^{kt}, \quad \text{where } k \text{ is a constant.}$$

Note: Continuous compounding $A = Pe^{rt}$ is one example of an exponential growth function.

Problem 7. * Suppose that \$1200 is invested at an interest rate of 2.75% per year, compounded continuously. After how many years will the initial investment be doubled? Do not round any intermediate computations, and round your answer to the nearest hundredth.

Problem 8. * Suppose that the number of bacteria in a certain population increases according to a continuous exponential growth model. A sample of 1800 bacteria selected from this population reached the size of 2115 bacteria in five hours. Find the hourly growth rate parameter.

Note: This is a continuous exponential growth model. Write your answer as a percentage. Do not round any intermediate computations, and round your percentage to the nearest hundredth.

Problem 9. * Suppose that \$2900 is initially invested in an account at a fixed interest rate, compounded continuously. Suppose also that, after two years, the amount of money in the account is \$3202. Find the interest rate per year.

Write your answer as a percentage. Do not round any intermediate computations, and round your percentage to the nearest hundredth.

Problem 10. *The mass of a radioactive substance follows a continuous exponential decay model, with a decay rate parameter of 3.8% per day. Find the half-life of this substance (that is, the time it takes for one-half the original amount in a given sample of this substance to decay).

Note: This is a continuous exponential decay model. Do not round any intermediate computations, and round your answer to the nearest hundredth.

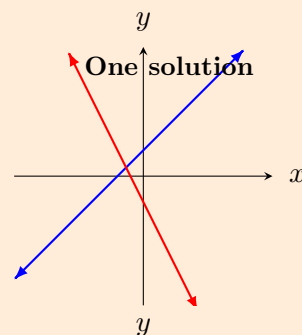
Linear Systems

Problem 11. *For each ordered pair, determine whether it is a solution to the system of equations.

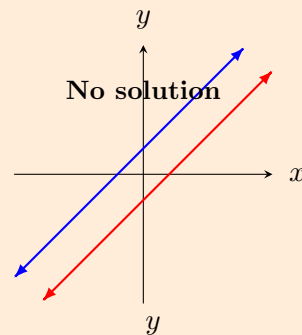
$$\begin{cases} 6x + 2y = 16 \\ y = -3x + 8 \end{cases}$$

(x, y)	Is a solution	Is not a solution
$(-5, 0)$		
$(1, 8)$		
$(-2, 14)$		
$(7, -13)$		

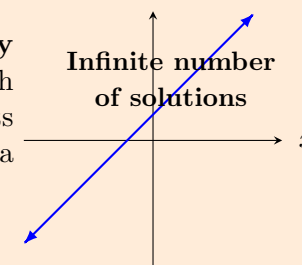
1. The graphs **intersect at exactly one point**, which gives the (single) ordered-pair solution of the system. Because the system has a solution, it is **consistent**. The equations are not equivalent, so they are **independent**.



2. The graphs **are parallel**, so there is **no solution**. Because the solution set is empty, the system is **inconsistent**. The equations are not equivalent, so they are **independent**.



3. The graphs **are the same line**, that is, they **coincide**. and there are an **infinite number of solutions**. The solution set is an infinite set of ordered pairs representing the points on the line. This type of system is **consistent** because there is a solution. The equations are equivalent, so they are **dependent**.



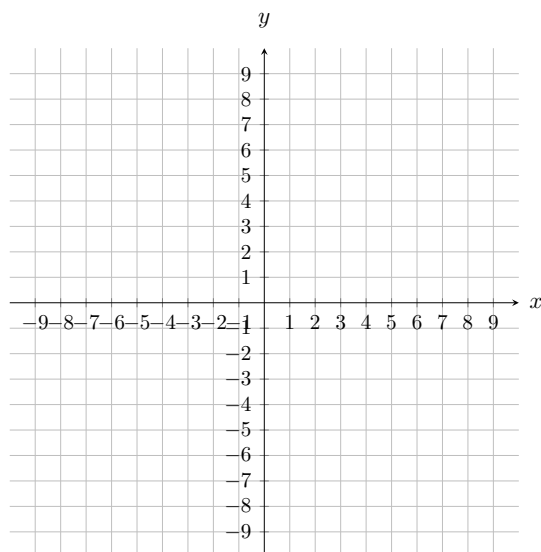
Problem 12. * Three systems of linear equations are shown. For each system, choose the best description of its solution. If the system has exactly one solution, give its solution.

System A	System B	System C
Line 1: $y = -\frac{5}{2}x + 4$	Line 1: $y = -x - 1$	Line 1: $y = -\frac{2}{3}x - 3$
Line 2: $5x + 2y = 8$	Line 2: $y = -\frac{1}{2}x - 2$	Line 2: $y = -\frac{2}{3}x$
☐ The system has exactly one solution. Solution: (,) ☐ The system has infinitely many solutions. ☐ The system has no solution.	☐ The system has exactly one solution. Solution: (,) ☐ The system has infinitely many solutions. ☐ The system has no solution.	☐ The system has exactly one solution. Solution: (,) ☐ The system has infinitely many solutions. ☐ The system has no solution.

Problem 13. * Here is a system of equations.

$$\begin{cases} y = -2x + 1 \\ y = 2x + 5 \end{cases}$$

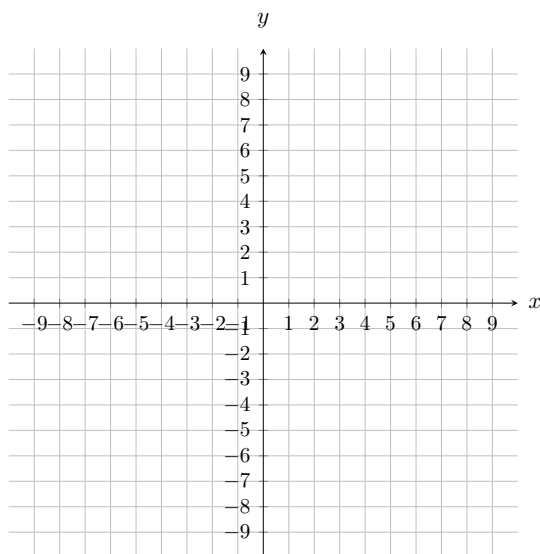
Graph the system. Then write its solution. Note that you can also answer "No solution" or "Infinitely many" solutions.



Problem 14. *Here is a system of equations.

$$\begin{cases} y = 2 - 2x \\ y = -2x + 2 \end{cases}$$

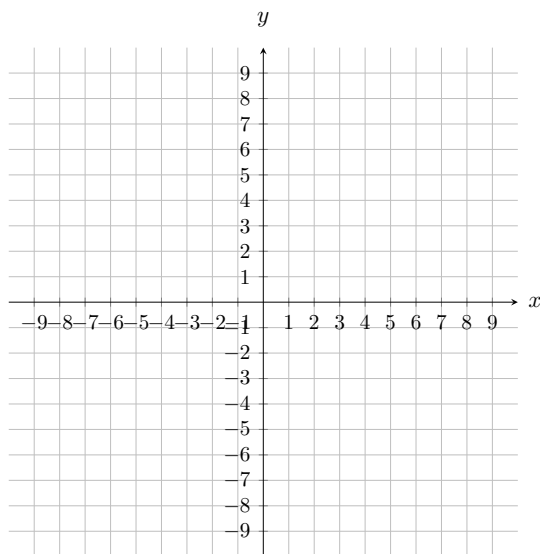
Graph the system. Then write its solution. Note that you can also answer "No solution" or "Infinitely many" solutions.



Problem 15. Here is a system of equations.

$$\begin{cases} y = -3x - 6 \\ y = -3x - 5 \end{cases}$$

Graph the system. Then write its solution. Note that you can also answer "No solution" or "Infinitely many" solutions.



Substitution Method

The substitution method involves using one equation to find an expression for one variable in terms of the other, and then substituting this expression into the other equation of the system.

Problem 16. Solve the system using the substitution method.

$$\begin{array}{rcl} 4x & - & 3y = 14 \\ x & - & 2y = 1 \end{array}$$

Elimination Method

To eliminate a variable, the coefficients of that variable in the two equations must be additive inverses. The three transformations that produce an equivalent system are listed here.

Transformations of a Linear System

1. Interchange any two equations of the system.
2. Multiply or divide any equation of the system by a nonzero real number.
3. Replace any equation of the system by the sum of that equation and a multiple of another equation in the system.

Problem 17. *Solve the system using the elimination method.

$$\begin{array}{rcl} 8x & + & y = 21 \\ -11x & - & y = -27 \end{array}$$

Problem 18. * Solve the system using the elimination method.

$$\begin{array}{rcl} 6x & + & 9y = -3 \\ 6x & + & 5y = 9 \end{array}$$

Problem 19. * Solve the system using the elimination method.

$$\begin{array}{rcl} -7x & - & 5y = 9 \\ -3x & - & 4y = 2 \end{array}$$

Problem 20. * Solve the system using the elimination method.

$$\begin{array}{rcl} -8x & + & 3y = -5 \\ 5x & - & 4y = 1 \end{array}$$

Problem 21. *Solve the system using a method of your choice.

$$\begin{array}{rcl} -\frac{1}{3}x & + & \frac{1}{6}y = 8 \\ \frac{3}{4}x & + & \frac{1}{2}y = -11 \end{array}$$

Problem 22. *Solve the system using a method of your choice.

$$\begin{array}{rcl} \frac{1}{6}x & - & \frac{1}{3}y = -4 \\ \frac{1}{2}x & + & \frac{3}{5}y = 4 \end{array}$$

Special Systems

If, while solving a system, both variables get eliminated then we can have the following possibilities:

Case 1. the resulting equation is false, and the system is inconsistent that is has no solutions.

Case 2. the resulting equation is true, and the system is consistent with infinitely many solutions.
Solutions set consists of ordered pairs satisfying the equivalent equations.

Problem 23. * For the system

$$\begin{array}{rcl} 7x & - & 5y = 1 \\ -7x & + & 5y = 2 \end{array} \text{ choose the best description of its solution.}$$

- (a) The system has no solution.
- (b) The system has a unique solution: $(x, y) = (\quad , \quad)$
- (c) The system has infinitely many solutions. They must satisfy the following equation: $y =$

Problem 24. * For the system

$$\begin{array}{rcl} 6x & - & y = -5 \\ -6x & + & y = 5 \end{array}$$

choose the best description of its solution.

- (a) The system has no solution.
- (b) The system has a unique solution: $(x, y) = (\quad , \quad)$
- (c) The system has infinitely many solutions. They must satisfy the following equation: $y =$

Problem 25. * For the system

$$\begin{array}{rcl} x + 2y & = & 2 \\ -x & = & 2y - 2 \end{array}$$

choose the best description of its solution.

- (a) The system has no solution.
- (b) The system has a unique solution: $(x, y) = (\quad , \quad)$
- (c) The system has infinitely many solutions. They must satisfy the following equation: $y =$